PRACTICE EXAM 2 OF 4 · EXAM STRATEGY - ALL DOMAINS

# Practice exam 2

Twenty new questions. Four pair with exam 1, where one changed detail changes the answer.

QUESTIONS 20 TIME About 30 minutes MODULE 33 of 35

admin@sthithapragna.com

> Quayside's production etcd cluster has five members.

**How many members can fail while etcd keeps accepting writes?**

A One

B Four

C Two

D Three

◆ PAUSE AND DECIDE

# How many members can fail while etcd keeps accepting writes?

A One

*Five members tolerate more than one.*

B Four

*That would leave one member. no majority.*

C Two

Quorum is three of five, so two may fail.

D Three

*That would leave two of five: no majority.*

THE SIGNAL

Quorum is a majority: floor(n/2) + 1.

> The web front end must be reachable from the internet at one stable address.
>
> The cluster runs on a public cloud, which should create and manage that address.

## Which Service type fits?

A LoadBalancer

B ClusterIP

C ExternalName

D A headless Service

◆ PAUSE AND DECIDE

## Which Service type fits?

A **LoadBalancer**
The cloud provisions one external address.

B **ClusterIP**
*Internal only, the default.*

C **ExternalName**
*An alias pointing outwards.*

D **A headless Service**
*No virtual IP; internal only.*

THE SIGNAL

Internet, one stable cloud address: LoadBalancer.

> The new GPU nodes must repel every Pod that does not tolerate them.
>
> Pods already running there without a toleration must also be moved off.

## What should be applied to the GPU nodes?

A A taint with the NoSchedule effect

B A taint with PreferNoSchedule

C A nodeSelector on the GPU Pods

D A taint with the NoExecute effect

◆ PAUSE AND DECIDE

## What should be applied to the GPU nodes?

A A taint with the NoSchedule effect
*Leaves running Pods in place.*

B A taint with PreferNoSchedule
*Only a preference; nothing is evicted.*

C A nodeSelector on the GPU Pods
*Attracts GPU Pods; repels nothing.*

D A taint with the NoExecute effect
Repels new Pods and evicts running ones.

THE SIGNAL

**Must evict running Pods too: NoExecute.**

> For a new booking API release, Quayside wants every user moved at the same moment.
>
> If anything goes wrong, it must switch back instantly. Double capacity for an hour is affordable.

## Which release strategy fits?

A A canary, starting small

B Blue-green, switching all at once

C Recreate, stopping old then new

D Shadow traffic, discarding replies

◆ PAUSE AND DECIDE

## Which release strategy fits?

A. A canary, starting small
*Moves users gradually, not at once.*

B. Blue-green, switching all at once
One switch forward, one switch back.

C. Recreate, stopping old then new
*Everyone at once, but with downtime.*

D. Shadow traffic, discarding replies
*Users never see the new version.*

THE SIGNAL

Everyone at once, instant rollback: blue-green.

> During an incident, an engineer needs each booking API Pod's node and IP address, one line per Pod.

## Which command gives that?

A kubectl get pods -l app=booking-api -o wide

B kubectl describe nodes

C kubectl top pods -l app=booking-api --containers

D kubectl get deploy booking-api -o yaml

◆ PAUSE AND DECIDE

## Which command gives that?

A kubectl get pods -l app=booking-api -o wide

Adds node and IP columns, one line per Pod.

B kubectl describe nodes

*Describes nodes, not where each Pod runs.*

C kubectl top pods -l app=booking-api --containers

*Shows CPU and memory use.*

D kubectl get deploy booking-api -o yaml

*Shows the Deployment, not its Pods.*

THE SIGNAL

**Node and IP per Pod: get pods -o wide.**

> New Pods show ImagePullBackOff. The events say: manifest for quayside/web:5.3.0 not found. The registry is public, and older tags pull fine.

## What is the most likely cause?

A The node has run out of CPU

B A liveness probe keeps failing

C The registry needs a pull secret

D Tag 5.3.0 is not in the registry

◆ PAUSE AND DECIDE

## What is the most likely cause?

A The node has run out of CPU
*CPU does not affect pulling.*

B A liveness probe keeps failing
*The container never started.*

C The registry needs a pull secret
*It is public, and older tags work.*

D Tag 5.3.0 is not in the registry
'Not found' for one tag: a missing or mistyped tag.

THE SIGNAL

'Manifest not found' on a public registry: the tag is wrong.

> The booking API runs 3 replicas. During node drains, at least 2 must always stay available.

## What enforces that during drains?

A A ResourceQuota on the namespace

B A LimitRange with default limits

C A PodDisruptionBudget, minAvailable 2

D A high PriorityClass for the Pods

◆ PAUSE AND DECIDE

## What enforces that during drains?

A A ResourceQuota on the namespace
*Caps namespace totals.*

B A LimitRange with default limits
*Sets per-container defaults.*

C A PodDisruptionBudget, minAvailable 2
Drain's evictions respect it.

D A high PriorityClass for the Pods
*Affects scheduling order, not evictions.*

THE SIGNAL

Availability during voluntary disruptions: PodDisruptionBudget.

> Quayside's dashboard shows: 99.7% of booking requests completed in under 300 ms over the last 28 days.

## What is that figure?

A An SLO

B An SLI

C An SLA

D An error budget

◆ PAUSE AND DECIDE

## What is that figure?

A An SLO
*A target, not a measurement.*

B An SLI
A measured indicator of service.

C An SLA
*A customer contract.*

D An error budget
*What an SLO leaves over.*

THE SIGNAL

A measured value: SLI. A target: SLO.

> The data-cleanup script from exam 1 must now run every night at 02:00, to completion each time.

## Which workload fits?

A A Job

B A CronJob

C A DaemonSet

D A Deployment

◆ PAUSE AND DECIDE

## Which workload fits?

A A Job
*Runs once; nothing repeats it.*

B A CronJob
Creates a Job on a schedule.

C A DaemonSet
*One copy per node, running forever.*

D A Deployment
*Restarts it after it finishes.*

THE SIGNAL

Finite work on a schedule: CronJob.

> Four web Pods on different nodes must all read and write the same folder of uploaded photos.

**Which access mode does the claim need?**

A ReadWriteOnce

B ReadOnlyMany

C ReadWriteMany

D ReadWriteOncePod

◆ PAUSE AND DECIDE

## Which access mode does the claim need?

A ReadWriteOnce
*One node at a time.*

B ReadOnlyMany
*Read-only.*

C ReadWriteMany
Many nodes, read-write.

D ReadWriteOncePod
*A single Pod only.*

THE SIGNAL

Many nodes, read-write: ReadWriteMany.

> An engineer is writing a Deployment manifest for a current cluster.

**Which apiVersion should it use?**

A apps/v1

B v1

C extensions/v1beta1

D batch/v1

◆ PAUSE AND DECIDE

## Which apiVersion should it use?

A apps/v1
Deployments live in the apps group.

B v1
*The core group: Pods, Services, ConfigMaps.*

C extensions/v1beta1
*Removed long ago.*

D batch/v1
*Jobs and CronJobs.*

THE SIGNAL

Deployments: apps/v1.

> The booking API's production settings live in a Kustomize overlay at overlays/prod.
>
> The engineer wants to build and apply it with kubectl alone.

## Which command does that?

A kubectl apply -f overlays/prod/kustomization.yaml

B helm install booking overlays/prod

C kubectl kustomize --apply overlays/prod

D kubectl apply -k overlays/prod

◆ PAUSE AND DECIDE

## Which command does that?

A kubectl apply -f overlays/prod/kustomization.yaml
*Applies the file itself, not the built overlay.*

B helm install booking overlays/prod
*Helm installs charts, not overlays.*

C kubectl kustomize --apply overlays/prod
*No such flag.*

D kubectl apply -k overlays/prod
-k builds the overlay and applies it.

THE SIGNAL

Kustomize with kubectl: apply -k.

> An auditor asks how Kubernetes stores Secret values in etcd, on a cluster with default settings.

**Which statement is accurate?**

A Encrypted with the cluster CA by default

B Hashed, so they cannot be read back

C Held only in memory, never in etcd

D Base64-encoded, but not encrypted

◆ PAUSE AND DECIDE

## Which statement is accurate?

A Encrypted with the cluster CA by default
*There is no default encryption.*

B Hashed, so they cannot be read back
*They can always be read back.*

C Held only in memory, never in etcd
*They are stored in etcd.*

D Base64-encoded, but not encrypted
Encryption at rest must be configured.

THE SIGNAL

**Default Secrets: encoded, not encrypted.**

> A container keeps restarting. describe shows Last State: Terminated, Reason: OOMKilled, Exit Code: 137.

## What happened?

A It exceeded its CPU limit

B It exceeded its memory limit

C Its liveness probe failed

D Its image could not be pulled

◆ PAUSE AND DECIDE

## What happened?

A It exceeded its CPU limit

*CPU over the limit is throttled, not killed.*

B It exceeded its memory limit

The kernel killed it at its memory limit.

C Its liveness probe failed

*That would not say OOMKilled.*

D Its image could not be pulled

*It started; pulls happen before.*

THE SIGNAL

**OOMKilled with 137: the memory limit.**

> A node joins a cluster, but its Pods get no IP addresses and cannot reach each other.

**Which component is responsible for Pod IPs and Pod-to-Pod connectivity?**

A kube-proxy

B CoreDNS

C The CNI network plugin

D The kubelet, on its own

◆ PAUSE AND DECIDE

# Which component is responsible for Pod IPs and Pod-to-Pod connectivity?

A kube-proxy
*Handles Service virtual IPs.*

B CoreDNS
*Answers DNS lookups.*

C The CNI network plugin
Assigns IPs and wires the Pod network.

D The kubelet, on its own
*Calls the runtime, which calls CNI.*

THE SIGNAL

**Pod IPs and the Pod network: CNI.**

> An engineer asks how the kubelet tells containerd to start a container.

## Which interface does it use?

A The CRI

B The CNI

C The CSI

D The OCI image spec

◆ PAUSE AND DECIDE

## Which interface does it use?

A **The CRI**
The kubelet's interface to runtimes.

B **The CNI**
*For network plugins.*

C **The CSI**
*For storage plugins.*

D **The OCI image spec**
*Describes image format, not the call.*

THE SIGNAL

**Kubelet to runtime: CRI.**

> An engineer uses kubectl to raise a Deployment's replicas in a cluster managed by Argo CD. Within a minute, Argo CD changes it back to match Git.

**Which Argo CD setting caused that?**

A prune

B targetRevision

C selfHeal

D A sync wave

◆ PAUSE AND DECIDE

## Which Argo CD setting caused that?

A prune
*Deletes objects removed from Git.*

B targetRevision
*Chooses which commit is tracked.*

C selfHeal
Reverts manual changes to match Git.

D A sync wave
*Orders resources during a sync.*

THE SIGNAL

**Manual change reverted: selfHeal.**

> After a default-deny egress policy is applied in the payments namespace, every outbound call fails with: could not resolve host.

## What is the fix?

A Restart the CoreDNS Pods

B Switch kube-proxy to nftables mode

C Remove the namespace's labels

D Allow egress to kube-dns on port 53

◆ PAUSE AND DECIDE

## What is the fix?

A Restart the CoreDNS Pods
*CoreDNS is fine; it is unreachable.*

B Switch kube-proxy to nftables mode
*Unrelated to name lookups.*

C Remove the namespace's labels
*Labels do not block DNS.*

D Allow egress to kube-dns on port 53
DNS traffic was blocked by the policy.

THE SIGNAL

Egress locked down and lookups fail: allow DNS.

> Quayside wants every container's logs collected centrally, with no changes to its applications.

## Which approach is standard?

A Apps write to a shared NFS volume

B A node agent DaemonSet, such as Fluent Bit

C A CronJob that dumps kubectl logs every hour

D Store the log lines in etcd

◆ PAUSE AND DECIDE

## Which approach is standard?

A Apps write to a shared NFS volume
*Needs app changes; fragile.*

B A node agent DaemonSet, such as Fluent Bit
Reads every container's logs on each node.

C A CronJob that dumps kubectl logs every hour
*Misses anything between runs.*

D Store the log lines in etcd
*etcd is not a log store.*

THE SIGNAL

Central logs, no app changes: a node-agent DaemonSet.

> Before the booking API starts, a database schema migration must run once and complete successfully.

## What should run the migration?

A An init container

B A native sidecar container

C An ephemeral container

D A second app container

◆ PAUSE AND DECIDE

## What should run the migration?

A An init container
Runs to completion before the app starts.

B A native sidecar container
*Keeps running beside the app.*

C An ephemeral container
*For debugging only.*

D A second app container
*Starts alongside the app, not before.*

THE SIGNAL

**Must finish before the app starts: init container.**

# Score yourself by domain

| DOMAIN | QUESTIONS | YOUR SCORE |
|---|---|---|
| Kubernetes Fundamentals (44%) | 1, 3, 5, 7, 9, 11, 14, 16, 20 | of 9 |
| Container Orchestration (28%) | 2, 6, 10, 13, 15, 18 | of 6 |
| Application Delivery (16%) | 4, 12, 17 | of 3 |
| Cloud Native Architecture (12%) | 8, 19 | of 2 |

ON THE EXAM

Compare with exam 1. A domain that is weak twice is where to spend your time.

# The pairs, and the constraint that changed

| THIS EXAM | EXAM 1 | WHAT CHANGED THE ANSWER |
|---|---|---|
| Q2 LoadBalancer | Q2 ClusterIP | Internet access with a cloud address |
| Q3 NoExecute | Q12 NoSchedule | Running Pods must also leave |
| Q9 CronJob | Q20 Job | It must repeat on a schedule |
| Q10 ReadWriteMany | Q10 ReadWriteOnce | Many nodes write, not one |

> ON THE EXAM
>
> If you got one of a pair wrong, you missed a constraint, not a fact. Reread the stem.

Practice exam 3: twenty more, with **more Delivery questions** to match the domain's weight.